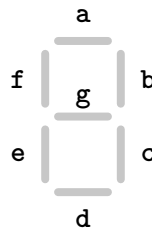
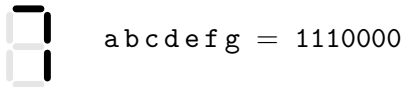


Problem 1.

A *seven-segment display* shows a digit by lighting up to seven little bars, labelled **a** to **g**. To draw a digit you switch some bars on and leave the rest off — so the whole display is really just **7 bits**: one bit per bar, where 1 means *on* and 0 means *off*. The seven bits for a digit are its row of the display's *truth table*.



Worked example — the digit 7. To make a **7** you light segments **a**, **b** and **c**; every other bar stays off.



Read the bars in order **a**, **b**, ..., **g** to get the seven output bits.

Your turn. For each digit below, shade the bars that light up, then write its seven output bits (1 = lit, 0 = off).

digit	display	a	b	c	d	e	f	g
0	8							
1	8							
2	8							
3	8							
4	8							
5	8							
6	8							
7	8							
8	8							
9	8							

Hint: it helps to picture the digit on the display first, then tick off each of the seven bars in turn.

Problem 2.

Advanced

A display chip can't read the shape "7" — it is handed the digit as a *binary number* and must switch on the right bars. (This is how a digital clock or a scoreboard lights up its digits.) Each digit arrives as binary on the *input*, and the seven segment bits come out as the *output*. So this time **both the input and the output are binary numbers**.

(a) Every digit must reach the display as a binary number. How many bits do you need to represent every digit from 0 to 9? _____

(b) Complete the truth table: write each digit as a 4-bit binary *input*, then the seven segment bits (a–g) it switches on as the *output*.

	input (binary)				output (segments)						
decimal	8	4	2	1	a	b	c	d	e	f	g
0											
1											
2											
3											
4											
5											
6											
7											
8											
9											